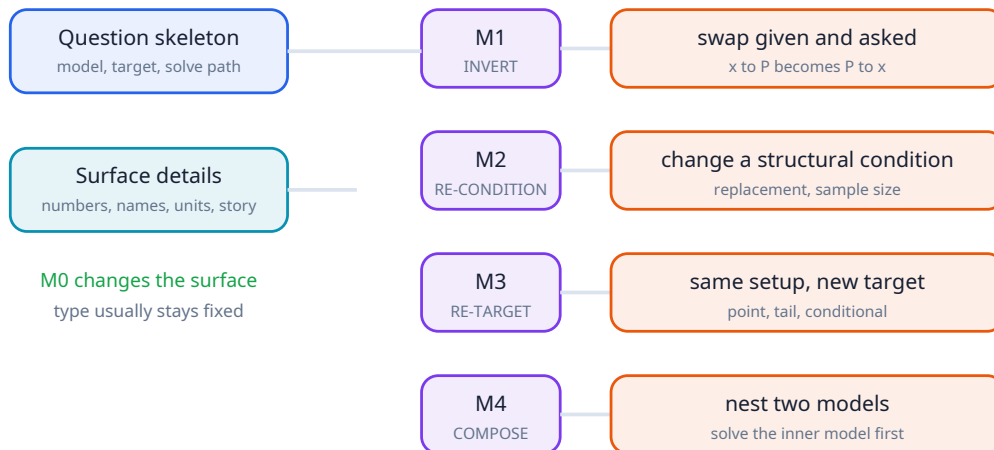


07 PATTERNS: how the setters mutate questions

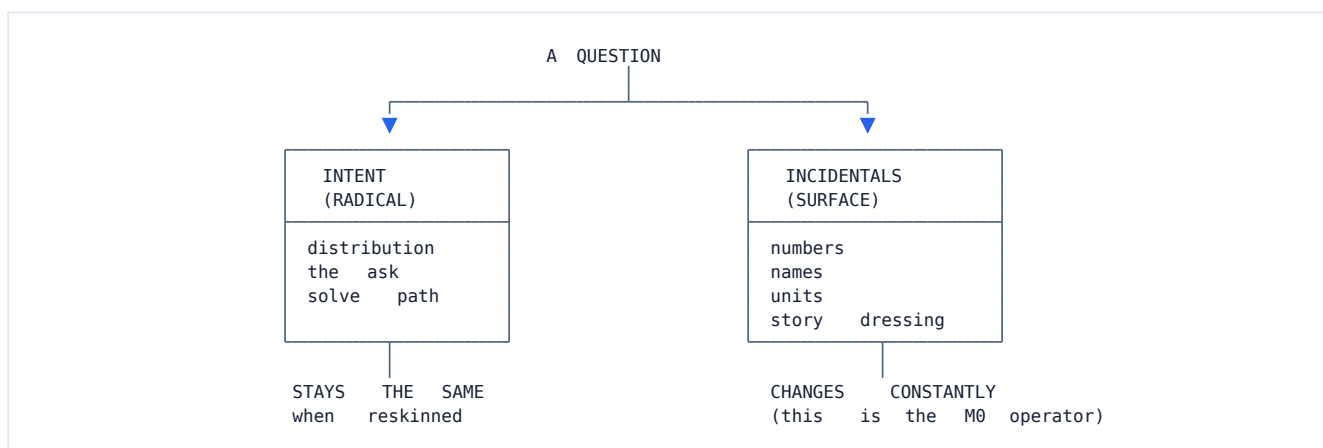
THE MUTATION TREE (intent vs incidentals)

Question mutation map



source match confidence belongs to the evidence ledger, not to visual similarity alone

Question mutation operators



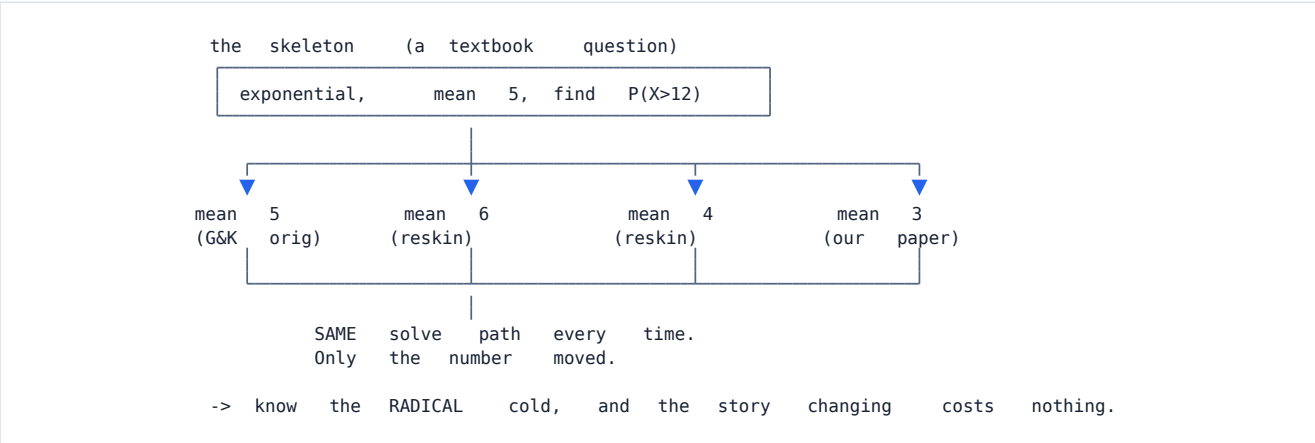
THE FIVE OPERATORS AS A TREE

M0 RE-SKIN | numbers/names/units/story
TYPE UNCHANGED | <- common in the reviewed papers; exact rate unmeasured
M1 INVERT | swap given <-> asked
NEW NODE | x->P becomes P->x
M2 RE-CONDITION | flip a structural condition
NEW NODE | replace/no-replace, n<30 vs n≥30
M3 RE-TARGET | same setup, different target
NEW NODE | P(=k)->P(≤k)->P(>k)->conditional
M4 COMPOSE | nest two models / chain
ALWAYS NEW | Poisson per min then binomial

THE NINE SLOTS AS A 3x3 GRID

1 POINT	2 TAIL	3 INTERVAL
P(X=k)	P(X>k)	P(a≤X≤b)
4 MOMENTS	5 PARAMS	6 INVERSE
E, Var	recover λ,p	x for given P
7 COUNT	8 CONDITIONAL 9 COMPOSE	
N x P	P(A B)	nest two
many distribution questions fit one cell. Foundations, Chebyshev, estimation,		
and mixed questions need their own method families.		

THE VALUE-DELTA IDEA, DRAWN



Many confirmed questions reuse textbook or circulating skeletons, often with changed values or stories. That does not prove that every question is copied, or that future questions must reuse known structures. This file records confirmed value changes, candidate mutation patterns, and nearby siblings worth practising.

1. The two layers of a question

INTENT	the deep structure: which distribution, which ask, which solve path.
This is the RADICAL. It does not change when the story changes.	
INCIDENTALS the numbers, names, units, and story dressing. These often leave the type fixed,	
but unit or support changes can alter the mathematics, so recheck the model.	

That single split is the whole system. When a question looks new, find its radical: if the radical is old, it is an old question wearing a costume.

2. The mutation operators

M0 RE-SKIN	change numbers, names, units, story. TYPE UNCHANGED.
pens become sensors become bulbs. Same math.	
examples: telephone mean 5 -> 6 -> 4 -> 3; subway 30min -> 4AM 15min	
M1 INVERT	swap given and asked. NEW NODE.
$x \rightarrow P(x)$ becomes $P \rightarrow x$. moments -> parameters.	
M2 RE-CONDITION flip a structural condition. NEW NODE when the formula changes.	
replacement on/off; $n < 30$ vs $n \geq 30$; discrete vs continuous.	
M3 RE-TARGET	same setup, different target. NEW NODE when the method differs.
$P(=k) \rightarrow P(\leq k) \rightarrow P(>k) \rightarrow$ conditional -> expected count.	
example: ABES asks SD, our paper asks VARIANCE on the same density.	
M4 COMPOSE	nest two models, or chain outputs. ALWAYS NEW NODE.
Poisson per-minute then binomial over the minute.	

3. The change table (does it create a new type or not)

change	new type?
numbers only	NO
names / units / story only	NO
rounding / decimal places	NO
same formula, different answer slot	NO
forward -> inverse	YES
point -> tail or interval	YES
unconditional -> conditional	YES
single -> expected count	YES
independence assumed -> not assumed	YES
with replacement -> without	YES
n small -> n large (if the distribution changes)	MAYBE
one model -> two models nested	YES
E -> Var -> $E[h(X)]$	YES
efficiency -> MSE comparison	YES

4. The value-delta log (mutations used on US, with the source)

telephone exponential	mean 5 (G&K ch5 ex10) -> 6 -> 4 -> 3	M0 x3
uniform Chebyshev	(-1,3) G&K ex15b -> (-1,1)	M0
subway/train wait	30-min subway G&K -> 15-min 4AM window	M0 + M2
Chebyshev inverse	sigma 2, 21/25 G&K -> sigma 3, 24/25	M0
machine life	"bread-making machine" -> "a machine"	M0
density $kx^3(4-x)^2$	ABES sample asks SD, our paper asks VAR	M3
pens defective	G&K family of 18 -> 12 pens, 3 parts	M0 (source OPEN)
Poisson $Y=2X$ variance	G&K answer 14 -> VERBATIM (uncostumed)	M0
t^2 biased estimator	G&K estimation ex6 -> VERBATIM	M0
six-coins 6400	G&K 7.26 family	M0
Poisson 9P/90P	G&K 7.30 family	M0
soldiers 68.22	G&K ch8 family	M0

Pattern in the currently confirmed source matches: M0 reskins dominate. M3 appears once in that confirmed set and M4 is less common. This is a study priority, not a claim about all 502 instances or a guarantee about a future paper.

5. The nine slots (the template every distribution shares)

slot 1	point	$P(X = k)$
slot 2	tail	$P(X > k)$, $P(X \geq k)$, complement
slot 3	interval	$P(a \leq X \leq b)$
slot 4	moments	E , Var , $E(X^2)$
slot 5	parameters	recover a parameter from a given probability or moment
slot 6	inverse	quantile / cutoff / min n
slot 7	expected count $N \times P$	
slot 8	conditional	$P(A B)$ inside the model
slot 9	composition	nest two models

These nine slots cover the main distribution questions. Probability foundations, Chebyshev, estimation, and mixed-model questions sit outside this grid. See 04-METHODS.md for those paths.

6. New type nodes the radical analysis generates (be ready for these)

7.a	DESIGN / MIN-SIZE INVERSION	"how large must n be so that ..."	HIGH RISK
binomial: solve n from the stated p and target probability			
CLT: solve n from the stated sigma and SE target			
Chebyshev: k for a given guarantee			
7.b	PARAMETER RECOVERY (general)	solve for l, p, mu, sigma from a probability or moment	
examples seen: $p=5/8$, $l=10$, $l=2.3026$, $\mu=87.18$, $b=5.5$, moment matching			
7.c	DERIVED-VARIABLE ALGEBRA	min/max/sum/difference of rvs	
two dice max $(2m-1)/36$, Poisson sum additivity,			
independent $B(n1,p)+B(n2,p)=B(n1+n2,p)$			
7.d	CONDITIONAL ON AN INEQUALITY	for $k \geq j$,	
$P(X=k X \geq j) = P(X=k)/P(X \geq j)$. The pens sibling gives 0.3207.			
7.e	DECISION / OPTIMISATION	choose the best option from an expectation	
magazine 3 vs 4 copies, capacity sizing, which estimator to use			
7.f	APPROXIMATION CHOICE	Poisson for binomial (asked), binomial $\rightarrow$ normal with a	
continuity idea ($z = 1.123$)			
7.g	MSE / BIAS-VARIANCE	$MSE(T4) = 25.75$ vs $Var(T3) = 0.3333$, the honest compare	
7.h	SHEET GAPS	geometric E and Var, hypergeometric mean: needed, not on	
the provided sheet			

7. Siblings one mutation away (not yet asked, risk-ranked)

rank	sibling	the mutation from something asked
1	normal one-unknown recovery	from the two-unknown system, drop one equation
2	binomial two-sided interval $P(2 \leq X \leq 4)$	from the one-sided tails (0.3367)
3	CLT for a SUM not a mean	sigma scales by $\sqrt{n}$
4	discrete conditional $P(X=k \mid X \geq j)$	from the continuous conditional
5	min-n design from stated p , sigma, and target	from parameter recovery
6	geometric variance / tail	from the geometric mean (graded once)
7	independence CHECK question	from the independence rules
8	MSE comparison	from the efficiency question with bias allowed
9	Poisson between-values	from the tail family
10	valid-cdf check	from the find-k normalisation

These are the questions most likely to appear that we have NOT drilled. If time is short, do the top four.

8. The recognition routine (apply to any unseen question)

1 read the story, ignore the numbers	
2 name the distribution (or "none", if it is event probability)	
3 name the slot (1 to 9 above)	
4 if a slot-5 or slot-6, it is an inversion: set up, solve, convert back	
5 if two distributions appear, it is slot 9: do the inner one first	
6 check the boundary word (at least / more than) and the support	
7 check the units	
8 then compute	